

Quantitative reconstruction from finite causal orders in a conformal diamond

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<https://github.com/Quanyra/quanyra-null-cone-lean>

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Abstract

Consider smooth volume densities on the unit square, bounded between $1/2$ and $3/2$, with uniform marginals and Euclidean Lipschitz constant at most two. In null coordinates these define a restricted class of $1 + 1$ -dimensional conformal diamonds with the same chronological relation. Observing only the partial orders induced by independent volume samples, we derive an explicit inverse estimate: the supremum difference of the densities, modulo exchange of the null axes, is at most $100(N^{-1/12} + \Delta_N)$, where Δ_N is the largest total variation distance between the finite order laws through size N . A logarithmic grid scale further gives $130((\log N/N)^{1/6} + \Delta_N)$ for every $N \geq 2$. The argument uses an occupied-grid forcing construction that constrains every two-order realizer, after one common axis exchange, to have interior rank error at most $30rn$. This deterministic finite theorem has been formalized in Lean 4. The concentration, cumulative reconstruction, density interpolation and proper-time arguments are given here as ordinary mathematical proofs and are not claimed as Lean verified. The constants are conservative, the exponents are not claimed optimal, and originality remains provisional. Public source and reproducibility artifacts support decentralized informal feedback and downstream testing.

Keywords: causal order; conformal geometry; two-dimensional posets; quantitative identifiability; formal verification.

Artifacts: Lean source, pinned dependencies, verification audit, and this manuscript are available in the public repository above.

Software archive: version 0.1.0, doi:10.5281/zenodo.23202763. This DOI identifies the software release including the original preprint; it is not a manuscript-specific DOI. The original release is preserved; this revision adds the improved grid-scale estimate.

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1 Introduction and scope

Chronology and volume have long motivated spacetime reconstruction. Janson’s equivalence theory for random posets supplies qualitative identification of order kernels from all finite laws [1]. Braun proves a smooth reconstruction theorem from chronological adjacency laws under geometric hypotheses in spacetime dimension at least three [2]. Henson gives a numerical construction of approximate flat embeddings in two dimensions [3]. These precedents establish the general reconstruction objective.

The present draft concerns an explicit finite inverse modulus for a fixed, regular, two-dimensional class. Its finite ingredient is approximate rigidity of *all* realizers, rather than an assumption of exact

unique realizability. Orientation forcing and modular decomposition are established methods [4]; Winkler's work on random two-dimensional orders is a particularly relevant comparison [5]. The full Winkler article has not yet been compared with this argument, so we make no certified priority claim.

The result concerns geometric inference within a stipulated class. It supplies neither new physical dynamics nor reconstruction of an arbitrary spacetime. No curvature estimate, practical sample-complexity claim, or societal carrying-capacity consequence is established.

2 Model and main estimate

Let $D = [0, 1]^2$, with null coordinates (u, v) , and set

$$g_\rho = -\rho(u, v)(du \otimes dv + dv \otimes du).$$

The interior of D is the spacetime; its closure is used for coefficient norms and endpoint limits. Its volume measure is $\rho du dv$, and chronology is

$$(u, v) \prec (u', v') \iff u < u', v < v'.$$

Let $\mathcal{K}$ consist of densities smooth on a neighborhood of D such that

$$\frac{1}{2} \leq \rho \leq \frac{3}{2}, \quad |\rho(x) - \rho(y)| \leq 2|x - y|_2, \quad \int_0^1 \rho(u, v) dv = \int_0^1 \rho(v, u) dv = 1 \quad (u \in [0, 1]). \quad (1)$$

In particular, total volume is one. Write $T(u, v) = (v, u)$ and $\rho^T = \rho \circ T$. The uniform-marginal gauge is fixed; the residual equivalence used here is axis exchange, not time reversal.

For k independent samples from $\rho du dv$, let $\nu_k(\rho)$ be the law of their labeled strict partial order. Labels are event indices, not coordinate rankings. For $N \geq 2$, define

$$\Delta_N(\rho, \sigma) = \max_{2 \leq k \leq N} \text{TV}(\nu_k(\rho), \nu_k(\sigma)), \quad d_{\text{conf}}(\rho, \sigma) = \min\{\|\rho - \sigma\|_\infty, \|\rho - \sigma^T\|_\infty\}.$$

Total variation is $\sup_A |\mu(A) - \nu(A)|$.

Theorem 2.1 (Inverse estimate). *For every $\rho, \sigma \in \mathcal{K}$ and integer $N \geq 2$,*

$$d_{\text{conf}}(\rho, \sigma) \leq 100(N^{-1/12} + \Delta_N(\rho, \sigma)). \quad (2)$$

The theorem is proved in Sections 4–5 using the finite theorem of Section 3. Only that finite component is currently formalized.

Corollary 2.2 (All-size identification). *If $\nu_k(\rho) = \nu_k(\sigma)$ for every k , then $\rho = \sigma$ or $\rho = \sigma^T$ on D .*

Proof. Every Δ_N vanishes. Let $N \rightarrow \infty$ in (2). The minimum of the two fixed nonnegative supremum norms is zero. $\square$

3 Occupied grids and finite rank rigidity

Take $n \geq 1$ points $x_i = (u_i, v_i) \in (0, 1)^2$, with distinct u coordinates and distinct v coordinates. Let $m \geq 16$, $r = 1/m$, and assume that no coordinate lies on a grid line and every open cell

$$(jr, (j+1)r) \times (kr, (k+1)r), \quad 0 \leq j, k < m,$$

contains a point. Put $J = [4r, 1 - 4r]^2$ and $B = \{i : x_i \notin J\}$. A realizer is a pair of strict total orders (L_1, L_2) whose intersection is the sampled chronology P . For any strict order L , set

$$\text{rank}_L(i) = 1 + \#\{j : L(j, i)\}.$$

Let $\text{rank}_u, \text{rank}_v$ denote the coordinate ranks. Define empirical marginals with inclusive thresholds, for example $F_{v,n}(t) = n^{-1} \#\{i : v_i \leq t\}$.

Theorem 3.1 (Finite realizer rank rigidity). *In addition to occupancy, suppose both empirical marginal errors on $[0, 1]$ are at most $2r$ and $\#B \leq 20rn$. For every realizer (L_1, L_2) there exists one exchange of these two orders, chosen for the entire sample, such that for every i with $x_i \in J$,*

$$|\text{rank}_{L_1}(i) - \text{rank}_u(i)| \leq 30rn, \quad |\text{rank}_{L_2}(i) - \text{rank}_v(i)| \leq 30rn.$$

Here L_1, L_2 denote the orders after the chosen exchange.

3.1 Orientation forcing

For a realizer define

$$Q(x, y) \iff L_1(x, y) \text{ and } L_2(y, x).$$

Every incomparable pair has precisely one Q orientation. The relation Q is transitive: two successive comparisons hold in L_1 , and their reverses hold in L_2 . Thus their endpoints are incomparable and have the corresponding Q comparison.

If the incomparability edges xy and xz share x , while y and z are comparable in P , their orientations at x agree. Otherwise a two-edge Q path would make y, z incomparable by transitivity. We use this elementary forcing rule for both the latent realizer and any proposed realizer.

3.2 A common anchor fixes separated pairs

Occupancy supplies points in the following cells:

$$\begin{aligned} A &\in (r, 2r) \times (1 - 2r, 1 - r), \\ B_0 &\in (1 - 2r, 1 - r) \times (r, 2r), \\ W &\in (1 - r, 1) \times (1 - 3r, 1 - 2r). \end{aligned}$$

The edges AB_0 and AW are incomparability edges and $B_0 \prec W$. Take any interior incomparable x, y with $u_x < u_y$ and $v_x - v_y \geq 3r$. The interval (v_y, v_x) contains an open grid interval: take $k = \lfloor v_y/r \rfloor + 1$; then $kr > v_y$ and $(k + 1)r \leq v_y + 2r < v_x$. Interior membership ensures an admissible row. Occupancy in its first-column cell gives

$$z \in (0, r) \times (kr, (k + 1)r).$$

The coordinate inequalities imply $z \prec x$, $z \prec A$, $y \prec W$, and $B_0 \prec W$. The sequence of incomparability edges

$$xy, \quad zy, \quad Ay, \quad AW, \quad AB_0$$

has comparable opposite endpoints at each successive shared vertex. Specifically the comparable pairs are (z, x) , (z, A) , (y, W) , and (B_0, W) . The forcing rule therefore gives

$$Q(x, y) \iff Q(z, y) \iff Q(A, y) \iff Q(A, W) \iff Q(A, B_0).$$

Exchanging L_1, L_2 reverses Q . Choose this exchange once so that $Q(A, B_0)$ agrees with the latent u order. All separated interior incomparable pairs then agree. Comparable pairs agree automatically in both realizers. No prime-graph or exact-uniqueness assumption is used.

3.3 Counting possible disagreements

For a fixed interior x_i , any disagreement in either order can only involve a point outside J or one with $|v_j - v_i| < 3r$. Set $a = v_i - 3r$ and $b = v_i + 3r$; both belong to $[0, 1]$. The strip is contained in $\{j : a < v_j \leq b\}$, so the marginal bound gives

$$\#\{j : |v_j - v_i| < 3r\}/n \leq F_{v,n}(b) - F_{v,n}(a) \leq (b - a) + 4r = 10r.$$

At most $20rn + 10rn$ partners can disagree. The absolute difference of two predecessor counts is at most the number of points on which their predecessor indicators differ. This proves both rank bounds, with the same exchange, and establishes Theorem 3.1.

4 Order-only reconstruction and probability

For a labeled dimension-two poset P , enumerate the finitely many pairs of total orders with intersection P and select the lexicographically first pair, using labels only to break ties. Define $A_n(P)$ as the empirical cumulative distribution function of

$$\left(\frac{\text{rank}_{L_1}(i)}{n}, \frac{\text{rank}_{L_2}(i)}{n} \right), \quad 1 \leq i \leq n.$$

For other posets define A_n arbitrarily. This is a single function of the observed order; grid points and anchors appear only in its accuracy proof. No computational efficiency is asserted.

Write $C_\rho(s, t) = \int_{[0,s] \times [0,t]} \rho$ and let C_n be the latent empirical cumulative function. Suppose occupancy holds and $|C_n - C_\rho| \leq r$ at every grid vertex. By monotonicity between vertices and the uniform population marginals:

1. each empirical marginal has error at most $2r$ at every threshold;
2. the full empirical cumulative error is at most $3r$;
3. the fraction outside J is at most $20r$, since each of the four marginal tails is bounded by $4r + r$.

Coordinate ties and grid-line points have probability zero. The finite theorem applies. Distinct coordinates and inclusive thresholds give the exact identity $\text{rank}_u(i)/n = F_{u,n}(u_i)$, and likewise for v ; there is no additional $1/n$ term. Each latent normalized rank differs from its coordinate by at most $2r$, so every interior reconstructed point is within $32r$ in each coordinate after one exchange.

Write q_i for the aligned reconstructed point, $b = \#B/n \leq 20r$, and $d = 32r$. For $s, t \in [0, 1]$ put $s_- = \max(0, s - d)$, $s_+ = \min(1, s + d)$ and similarly for t . Interior-point indicator inclusions give, simultaneously for all thresholds,

$$C_n(s_-, t_-) - b \leq A_n^S(s, t) \leq C_n(s_+, t_+) + b.$$

If a lower threshold clips to zero, its latent rectangle contains no points because latent coordinates are strictly positive. Upper clipping uses the unit-square bounds. This includes reconstructed ranks equal to one and values $d > 1$. Uniform population marginals bound the mass change from moving the two thresholds by $2d = 64r$; the latent empirical replacement costs $3r$. Each direction uses one boundary penalty b . Consequently

$$\min\{\|A_n(P) - C_\rho\|_\infty, \|A_n(P) - C_{\rho^T}\|_\infty\} \leq 87r. \quad (3)$$

The transpose of a cumulative function exchanges its arguments and preserves the supremum norm.

For $n \geq 16^4$ choose $m = \lfloor n^{1/4} \rfloor$. Every cell has mass at least $1/(2m^2)$, giving an occupancy failure bound $m^2 \exp(-n/(2m^2))$. At each grid vertex, the empirical mass is an average of independent Bernoulli variables. Hoeffding's inequality [6] and a union bound give failure probability $2(m+1)^2 \exp(-2n/m^2)$ for vertex error at most r . Independence across vertices or overlapping suborders is not assumed. Since $n \geq m^4$ and $e^t \geq t^2/2$,

$$\begin{aligned} m^2 e^{-n/(2m^2)} + 2(m+1)^2 e^{-2n/m^2} &\leq m^2 e^{-m^2/2} + 2(m+1)^2 e^{-2m^2} \\ &\leq \frac{8}{m^2} + \frac{(m+1)^2}{m^4} \leq \frac{12}{m^2} \leq \frac{12}{256} < \frac{1}{10}. \end{aligned}$$

Also $m \geq n^{1/4}/2$. Thus (3) holds with probability at least 9/10, uniformly in $\rho \in \mathcal{K}$, with

$$a_n = 174n^{-1/4} \tag{4}$$

in place of $87r$.

5 From finite laws to coefficient error

Lemma 5.1 (Cumulative interpolation). *For $\rho, \sigma \in \mathcal{K}$ and $\epsilon = \|C_\rho - C_\sigma\|_\infty$,*

$$\|\rho - \sigma\|_\infty^3 \leq 2048\epsilon.$$

Proof. Put $f = \rho - \sigma$ and $\delta = \|f\|_\infty \leq 1$. The function f is 4-Lipschitz. If $\delta > 0$, take a maximizer x of $|f|$ and $h = \delta/16$. In each coordinate choose a length- h interval extending from x towards the half of $[0, 1]$ with more room. Their product R lies in D . For $z \in R$, $|z - x|_2 \leq 2h$, so $|f(z) - f(x)| \leq 8h = \delta/2$. The sign is fixed and

$$\left| \int_R f \right| \geq (\delta/2)h^2 = \delta^3/512.$$

Inclusion-exclusion expresses this integral by four corner values of $C_\rho - C_\sigma$, of total absolute value at most 4ϵ . The result follows; $\delta = 0$ is immediate. $\square$

Let

$$\epsilon_* = \min\{\|C_\rho - C_\sigma\|_\infty, \|C_\rho - C_{\sigma^T}\|_\infty\}, \quad t_n = \text{TV}(\nu_n(\rho), \nu_n(\sigma)).$$

Transposition is a supremum-norm isometry, so the minimum of the four cross-distances between the two transpose orbits is exactly ϵ_* . The selector has a finite order domain; every output preimage used below is therefore measurable without a function-space measurability assumption. If $\epsilon_* > 2a_n$, the sets of outputs within a_n of the two transpose orbits are disjoint. Each law places at least 9/10 of its mass on its own good set and at most 1/10 on the other's. Taking the preimage of a good set under the same order-only map A_n yields $t_n \geq 4/5$. Therefore $t_n < 4/5$ implies $\epsilon_* \leq 2a_n$. Applying Lemma 5.1 to the minimizing orientation yields $d_{\text{conf}} \leq (4096a_n)^{1/3}$. If $t_n \geq 4/5$, the elementary $d_{\text{conf}} \leq 1$ gives $d_{\text{conf}} \leq (5/4)t_n$. In both cases

$$d_{\text{conf}} \leq (4096a_n)^{1/3} + \frac{5}{4}t_n.$$

For $n \geq 65536$, substitute (4) and use $4096 \cdot 174 = 712704 < 100^3$. Setting $n = N$ and using $t_N \leq \Delta_N$ proves (2). For $2 \leq N < 65536$, $d_{\text{conf}} \leq 1 < 100N^{-1/12}$ proves it directly.

Remark 5.2 (Magnitude and optimality). The bare N term becomes smaller than one only when $N > 10^{24}$. The bound establishes a polynomial modulus, not a useful finite sample budget. No optimality of the $N^{-1/12}$ rate is asserted.

5.1 A logarithmic grid scale

Theorem 5.3 (Improved inverse estimate). *For every $\rho, \sigma \in \mathcal{K}$ and integer $N \geq 2$,*

$$d_{\text{conf}}(\rho, \sigma) \leq 130((\log N/N)^{1/6} + \Delta_N(\rho, \sigma)).$$

The logarithm is natural. The observable, class and residual quotient are those of Theorem 2.1.

Proof. For $n \geq 65536$ set $x = \sqrt{n/(8 \log n)}$, $m = \lfloor x \rfloor$, and $r = 1/m$. The function $t/\log t$ is increasing for $t > e$, since its derivative is $(\log t - 1)/(\log t)^2$. At the cutoff, $\log 65536 = 16 \log 2 < 16$, hence $x^2 > 512 > 16^2$. Thus $m \geq 16$ and $\log n \geq 1$. Also $m \geq x - 1 \geq x/2$, so

$$m^2 \leq \frac{n}{8 \log n}, \quad r \leq 2\sqrt{\frac{8 \log n}{n}}.$$

The deterministic reconstruction proof is unchanged. The two failure terms in Section 4 satisfy

$$\begin{aligned} m^2 e^{-n/(2m^2)} &\leq \frac{1}{8n^3 \log n}, \\ 2(m+1)^2 e^{-2n/m^2} &\leq \frac{n^{-15}}{\log n}. \end{aligned}$$

Indeed the exponents are at least $4 \log n$ and $16 \log n$; $(m+1)^2 \leq 4m^2$ supplies the second prefactor. Their sum is at most $1/(8n^3) + n^{-15} < 1/10$. Consequently success is at least $9/10$, with cumulative error

$$a_n = 174\sqrt{8}\sqrt{\log n/n}.$$

The same orbit-separation and interpolation argument gives

$$d_{\text{conf}} \leq (712704\sqrt{8})^{1/3}(\log n/n)^{1/6} + \frac{5}{4}t_n.$$

The coefficient is below 130: squaring the positive cubed comparison reduces it to

$$8 \cdot 712704^2 = 4063575932928 < 4826809000000 = 130^6.$$

Set $n = N$ and use $t_N \leq \Delta_N$.

For $2 \leq N < 65536$, $\log 2 \geq 1/2$ follows by integrating $1/t$ on $[1, 2]$. Thus $\log N/N \geq 1/131072 > (1/8)^6$, and the displayed N term exceeds $130/8 > 1 \geq d_{\text{conf}}$. This proves the small- N branch without a grid or concentration assertion there. $\square$

The new estimate improves the asymptotic N term but is not an optimal rate or a practical sampling guarantee. It uses the same prose probability and continuum bridges, which remain outside the current Lean verification scope.

6 Observable labels and proper time

Proposition 6.1 (Labeled and unlabeled laws). *For each k , the total variation distance of the labeled order laws equals that of their unlabeled order-isomorphism laws.*

Proof. Exchangeability makes each labeled poset in a relabeling orbit O have probability $p_O/|O|$. For the second law let the orbit mass be q_O . Then labeled total variation is

$$\frac{1}{2} \sum_O \sum_{P \in O} |p_O/|O| - q_O/|O|| = \frac{1}{2} \sum_O |p_O - q_O|,$$

the unlabeled distance. Order duality is not included in relabeling. $\square$

The label-based tie-breaking in A_n therefore does not confer extra information in these iid law comparisons.

For an absolutely continuous future-directed curve $\gamma = (u, v)$, with nonnegative coordinate derivatives almost everywhere, define

$$L_\rho(\gamma) = \int \sqrt{2\rho(\gamma)u'v'}.$$

Time separation is the supremum over such curves between the endpoints, and is zero if no such future curve exists.

Proposition 6.2 (Proper-time comparison). *There is one choice $S \in \{\text{id}, T\}$ such that, for all endpoints $p, q \in D$,*

$$|\tau_\rho(p, q) - \tau_\sigma(Sp, Sq)| \leq d_{\text{conf}}(\rho, \sigma).$$

Proof. First set $\delta = \|\rho - \sigma\|_\infty$. The density floor gives

$$|\sqrt{2\rho} - \sqrt{2\sigma}| = \frac{2|\rho - \sigma|}{\sqrt{2\rho} + \sqrt{2\sigma}} \leq \delta.$$

For a common monotone curve from p to q , Cauchy–Schwarz gives $\int \sqrt{u'v'} \leq \sqrt{(q_u - p_u)(q_v - p_v)} \leq 1$. Thus curve lengths differ by at most δ . Both suprema are finite, since every such length is at most $\sqrt{3}$, and the curve classes agree. Taking suprema proves the identity-aligned bound. Axis exchange preserves the product of derivatives and maps endpoints and curves consistently. Choose the orientation attaining d_{conf} . $\square$

7 Formal verification and reproducibility

The finite result appears in `QuanyraNullCone/Bridges.lean` as `finite_realizer_rank_rigidity_specified`. Its existential Boolean exchange precedes the universal event quantifier and is shared by both bounds. The stronger general theorem needs only the v marginal assumption, alongside occupancy, the boundary count and coordinate injectivity; the specified specialization retains all the hypotheses stated here.

The formal proof comprises the transitive incomparability orientation, shared-vertex forcing, occupied-grid witnesses, common orientation, strip/disagreement counts, and the two rank inequalities. Lean 4.30.0 and mathlib commit `c5ea00351c28e24afc9f0f84379aa41082b1188f` are pinned. Both final theorem audits report only `propext`, `Classical.choice`, and `Quot.sound`; no additional axiom or admitted proof is used. The abstract forcing lemmas report no axioms. The proof sources contain no `sorry` or `admit`.

The root library and `checks/Audit.lean` compile successfully. The probability-to-density theorem, all-size corollary, label-law equivalence and proper-time comparison in this manuscript remain outside the current formalization. Kernel checking of the finite theorem is not peer review or originality certification.

Two supplementary falsification scripts check 36 exact-coordinate grids with 865445 separated edges, and all 120 five-event coordinate permutations with 771 valid realizers. They are sanity checks, not substitutes for the universal Lean theorem. Instructions and the statement-to-source mapping are in the repository’s `INTEGRITY.md` and verification notes.

8 Discussion and research provenance

The proof uses a fixed gauge and uniform positive density to separate small geometric ambiguity from large-scale orientation. Extending the class, sharpening the grid scale, controlling derivatives, and formalizing the continuum bridges are separate questions. This draft does not claim to settle the general causal-set Hauptvermutung.

The work was developed with OpenAI Codex assistance under Daniel Eric Fredriksen’s research direction. AI-assisted derivation and writing are disclosed. The open-source workflow supports decentralized informal feedback, reuse and downstream testing; formal specialist review is not a publication prerequisite. The repository retains the derivation, exact-source comparisons and verification provenance. The candidate contribution is the explicit restricted inverse estimate and its approximate realizer-rigidity ingredient; publication-level originality assessment remains open.

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